Project Design Phase
Problem – Solution Fit Template

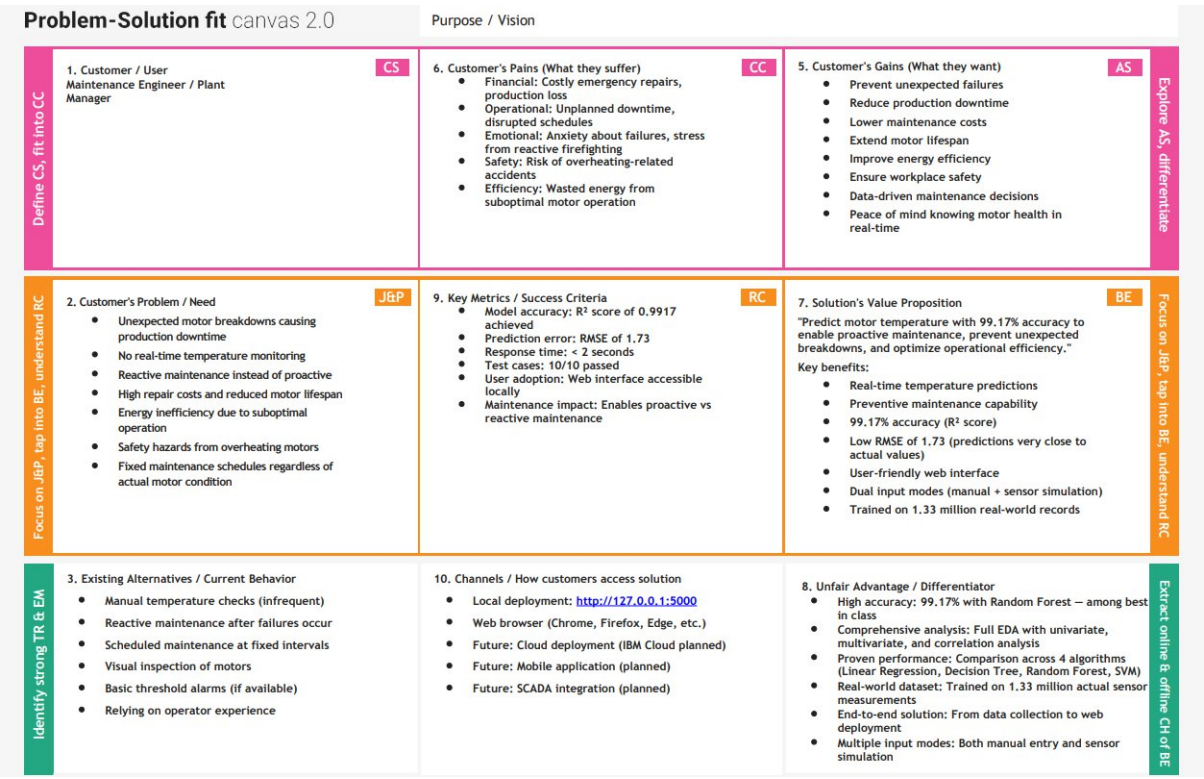
Date	14 February 2025
Team ID	LTVIP2026TMIDS81934
Project Name	Electric Motor Temperature Prediction using Machine Learning
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**



4. Solution Concept

A machine learning-based web application that predicts permanent magnet synchronous motor (PMSM) temperature in real-time using 7 operational parameters.

Key features:

- Random Forest algorithm with 99.17% accuracy
- 7 input parameters: ambient temperature, coolant temperature, voltage (u_d, u_q), motor speed, current (i_d, i_q)
- Web interface with manual input and sensor simulation modes
- Real-time predictions (response < 2 seconds)
- Local deployment capability

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